

Jingfan Tang

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Education

Shanghai Jiao Tong University

B.E. in Electrical & Computer Engineering, GPA: 3.83/4.0

Sept 2023 – Aug 2027
(expected)

Carnegie Mellon University

Department of Electrical & Computer Engineering, College of Engineering

Jan 2026 – May 2026

Selected Projects

Multi-robot Routing with Probabilistic Terminals for Cooperative Target Search

Yunpeng Lyu, *Jingfan Tang*, Zhongqiang Ren

In submission to IEEE International Conference on Robotics and Automation (ICRA), 2027

Goal2Pixel: Grounding Goals to Pixels for Vision-Language Navigation

Muyi Bao, Yuxin Cai, Hang Xu, Zongtai Li, Jinxi He, *Jingfan Tang*, Chen lv, Ji Zhang, Yaqi Xie, Wenshan Wang

In submission to IEEE International Conference on Robotics and Automation (ICRA), 2027

RPV*: Routing with Probabilistic Vertices and Its Use in Autonomous Exploration

Jingfan Tang, Ruizhi Deng, Yunpeng Lyu, Shizhe Zhao, Zhongqiang Ren

International Symposium of Robotics Research (ISRR), 2026

HEHA: Bounded Suboptimal Planning for Constrained Routing and Its Use in Heterogeneous Multi-Robot Exploration of Unknown Environments

Longrui Yan[†], Yiyu Wang[†], *Jingfan Tang*[†], Yunpeng Lyu, Shizhe Zhao, Chao Cao, Zhongqiang Ren

In submission to IEEE Transactions on Robotics (T-RO), 2026

Research Experience

Undergraduate Researcher, Shanghai Jiao Tong University

Mar 2025 – Present

Project: Fast Search Algorithms for Multi-Objective Traveling Salesman Problems

- Developed an exact search framework for multi-objective TSP that efficiently enumerates Pareto-optimal tours under multiple, potentially conflicting edge-cost objectives.
- Leveraged a new **Pareto-set maintenance mechanism** for efficiently storing, querying, and updating non-dominated search labels, significantly reducing the overhead of repeated dominance checks and enabling stronger pruning during search.

Project: Hierarchical Autonomous Exploration System with Map Prediction

- Developed a prediction-aware autonomous exploration system that combines learned map prediction with hierarchical global/local planning, improving exploration efficiency by **7–20%** over state-of-the-art baselines.
- Formulated **Routing with Probabilistic Vertices (RPV)** to explicitly reason about uncertain predicted regions, and developed optimal and bounded-suboptimal search algorithms with theoretical guarantees.
- Deployed and validated the system on an AgileX Scout Mini robot in cluttered indoor and outdoor environments, demonstrating reliable sim-to-real transfer.

Project: Hierarchical Multi-Robot Exploration System for Heterogeneous Agents

- Studied multi-robot exploration under heterogeneous mobility and sensing constraints, where robots have different feasible routes and traversal costs.
- Developed search-based routing and task-allocation methods with bounded-suboptimality guarantees, and integrated them into a hierarchical exploration system.
- Evaluated the system in large-scale simulated environments and real robotic platforms, demonstrating improved exploration efficiency and robustness over existing exploration methods.

Advisor: Prof. Zhongqiang Ren

Research Intern, Carnegie Mellon University

Jan 2026 – Present

Project: Language-Instructed Object Navigation Benchmark

- Developed a benchmark for long-horizon embodied navigation that combines coarse-grained language instruction following with downstream object-goal navigation.
- Constructed staged navigation tasks by processing landmark-based trajectories and sub-instructions, including extraction, filtering, restructuring, and difficulty control based on semantic and visibility cues.
- Built rollout and evaluation pipelines to analyze language grounding, intermediate navigation behavior, and subsequent object-search efficiency across different task settings.

Advisor: Prof. Ji Zhang

Skills

- **Languages:** C, C++, Python, Matlab
- **Framework and Operating System:** Linux, Git, ROS (Robot Operating System)
- **Speaks:** Mandarin, English (TOEFL 107)

Honor and Awards

- **The John Wu & Jane Sun Merit Scholarship (Top 2% in overall performance)**, Shanghai Jiao Tong University, 2025
- **Undergraduate Excellent Scholarship (Top 5% in academic performance)**, Shanghai Jiao Tong University, 2024–2025
- **The Yu Liming Scholarship (Top 5% in overall performance)**, Shanghai Jiao Tong University, 2024

Services

- **Teaching Assistant of Course MATH2030J Discrete Mathematics**
Lecturer: Dr. Runze Cai.
- **Teaching Assistant of Course MATH2550J Honor Multivariate Calculus**
Lecturer: Dr. Runze Cai.
- **Teaching Assistant of Course ECE2810J Data Structure and Algorithms**
Lecturer: Dr. Weikang Qian.
- **Organizer of CMU-VLN-Challenge 2026**
- **Reviewer for IROS 2026, Autonomous Robots 2026**
- **Leader and Advisor of GC Student Advising Center**
Conducted Workshops on graduate school application (2025 Summer), and on freshman course selection (2024 Fall, 2025 Fall).